MIL-D-0038134B(AS)
1 April 1974
Used in Lieu of
MIL-D-38134A(USAF)
20 May 1971

MILITARY SPECIFICATION

DETECTOR, MAGNETIC AZIMUTH DSU-4A/A, HIGH TEMPERATURE

"This limited coordination military specification has been prepared by the Naval Air Systems Command, Department of the Navy based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-D-38134A (USAF). It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement."

1. SCOPE

1.1 Scope - The equipment covered by this specification shall detect the direction of the earth's magnetic field with respect to the direction of the aircraft and shall be capable of operating with ambient temperatures up to 200°C.

1.2 Classification - The detector covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Transmitter, Induction TRU-79A/A	3.5.1
Compensator, Electronic MXU-394/A	3.5.2

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.5.

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-S-4473	Shielding, Magnetron Tubes and Magnets for Air, Shipment
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-W-22759	Wire, Electric, Fluorocarbon - Insulated, Copper and Copper Alloy
MIL-C-25624	Calibrator Set, Magnetic Compass, Type A/E37T-10

Naval Air Systems Command

AR-34	Failure classification for Reliability Testing, General Requirements for

Standards

Military

MIL-STD-100	Engineering Drawing Practices

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MIL-STD-130	Identification marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-765	General Requirements for Aircraft Compass Swinging
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipment)
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS24264	Connector, Receptacle, Electrical Flange Mount, Miniature
MS24266	Connector, Plug, Electrical, Straight, Miniature

DOD Document

SD-6	Provisions Governing Qualification

2.1.1 Availability of Documents - When requesting specification, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Qualification - This specification makes provision for Qualification Testing in accordance with the requirements of SD-6 (See 6.8).

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400. Specifications for items not covered herein shall be selected in accordance with MIL-STD-143.

(b) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity.

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

3.2.2 Aluminum Alloy Parts - Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

(a) Small holes and case inserts need not be anodized.

(b) Castings that will contain non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing prior to installation of the inserts.

(c) Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.2.3 Non-Magnetic Materials - Non-magnetic materials shall be used for all parts of the detector except where magnetic materials are essential.

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3.3 Design and Construction - The detector shall conform with all applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the detector excluding cables, shall be a minimum consistent with good design and shall not exceed 36.2 ounces (1031 gm).

Component weights shall be as follows:

Transmitter	28.8 ounces (818 gm) maximum
Compensator	7.5 ounces (213 gm) maximum

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide.

3.3.2.1 Operational Stability - The detector shall operate with satisfactory performance, continuously or intermittently, for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - The detector, including any Built-In-Test provisions, shall have a specified mean (operating) time between failures of 1500 hours when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Failure - Failure shall be considered as leakage or any malfunction which results in the inability of detector to perform its required function within established limits (careless handling or involuntary destruction excepted).

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The detector shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - Interconnections between the compensator and transmitter shall be as shown in Figure 1 and the detector shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088.

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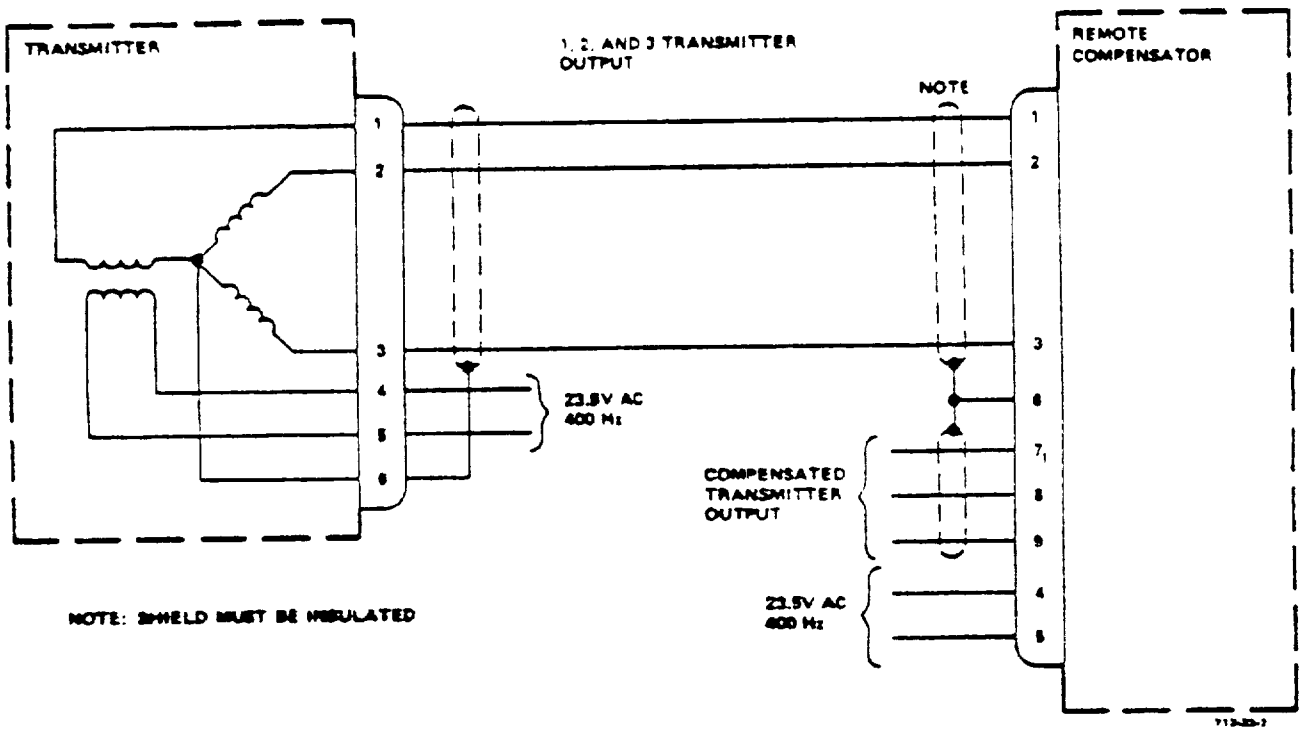


Figure 1
Interconnections

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3.3.4 Interchangeability - The detector shall meet the interchangeability requirements of MIL-E-5400.

3.3.4.1 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment or system supplied or designated by the procuring activity.

3.3.5 Maintainability

3.3.5.1 Organizational Level Maintainability - The magnetic azimuth detector consisting of a transmitter and compensator shall be designed and built for easy and simple installation, removal and replacement in an aircraft. After verification of detector failure the maximum removal and replacement time, excluding aircraft access time for either component shall be 30 minutes not including system calibration time. If swinging the aircraft is necessary following a replacement of any detector component, this time shall not be included in the specified removal and replacement time.

3.3.5.2 Intermediate Level Maintenance - The compensator shall be designed and built such that 90% of the test and repair shall not exceed:

Test time to verify failure	30 minutes max
Repair time	120 minutes max
Mean repair time	75 minutes
Retest time	30 minutes max

3.3.5.3 Depot Level Maintenance - The transmitter shall be designed and built such that 90% of the test and repair times shall not exceed:

Transmitter

Test time to verify failure	90 minutes max
Repair time including unseal and reseal	360 minutes max
Repair time excluding unseal and reseal	180 minutes mean
Retest time to confirm operational fitness	180 minutes max

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3.3.6 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient 25 °C ± 10 °C
Altitude	Normal Ground
Vibration	25 to 35 Hz, 0.002 inch (0.05 mm) to 0.005 inch (0.13 mm) circular vibration
Humidity	Room ambient up to 90% relative humidity
Input power voltage	23.5 ± 0.5v, 400 ± 5 Hz
Field Strength	Horizontal field 0.18 ± 0.02 oersted Vertical field 0.54 ± 0.02 oersted

3.3.8 Service Conditions - The detector shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specifications MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.8.1 Voltage and Frequency Variation - A shift in the input voltage and frequency to any combination between 22 and 25V and 360 and 440 Hz shall not cause the heading signal to shift more than ±.5° at any point on the azimuth.

3.3.8.2 High Temperature - The transmitter shall be capable of operating continuously in ambient temperatures up to 200 °C, and the compensator shall be capable of operating continuously in ambient temperatures up to 71 °C.

3.3.8.3 Low Temperature - The detector shall be capable of operating continuously in ambient temperatures down to -55 °C.

3.3.8.4 Vibration - The detector shall be capable of operating as specified herein under all vibratory conditions incident to normal use with not more than ±1° signal error.

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3.3.8.5 Altitude - The detector shall operate at altitudes up to 80,000 feet at any operational temperature range and within the tolerances for the ranges specified herein.

3.3.8.6 Acceleration - The detector shall withstand an acceleration of 3g during continuous operation and the following acceleeations in the directions specified:

(a) Fore: 9.0g

(b) Aft: 3.0g

(c) Up: 4.5g

(d) Down: 13.5g

(e) Lateral: 6.0g

3.3.8.7 Fungus - The detector shall be capable of operating as specified herein after subjection to fungus as encountered in tropical climates.

3.3.8.8 Dust - The detector shall be capable of operating as specified herein after subjection to sand and dust particles as encountered in desert areas.

3.3.8.9 Humidity - The detector shall be capable of operating as specified herein after subjection to relative humidity up to 90 percent.

3.3.8.10 Salt Fog - The detector shall be capable of operating as specified herein after subjection to salt sea atmosphere for 48 hours.

3.3.8.11 Shock - The detector shall be capable of operating as specified herein after subjection to shock of 15g for 11 milliseconds.

3.3.8.12 Explosive Atmosphere - When operated in an explosive atmosphere, the compensator shall not cause an explosion.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from performance under standard conditions will be specified.

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3.4.1 Operation - The detector shall provide a means of detecting the direction of the earth's magnetic field with respect to the direction of the aircraft and provide a 800 Hz output data signal for aircraft utilization.

3.4.2 Calibration Requirement - The detector shall be suitable for use with the A/E37-10 Magnetic Compass Calibrator Set (MC-1) conforming to MIL-C-26524, the Magnetic Compass Calibrator Set (MC-2) Sperry Flight Systems, Part No. 2592080-5, and the Magnetic Compass Calibrator Set (MC-100) Sperry flight Systems, Part No. 4021608-901.

3.5 Detail Requirements

3.5.1 Transmitter - A transmitter assembly shall consist of an interchangeable pendulous mechanism sensor and a pre-indexed mounting ring. The transmitter with the pre-indexed mounting ring shall conform to Figures 2 and 2A. When properly energized in the earth's magnetic field, the transmitter shall provide the indicator signal transformer with the proper direction signal for the compass system. The three secondary signal coils shall be internally Y-connected with the neutral point made externally available (see Figure 1). The secondary coils leg-to-leg inductance when measured at 1,000 cycles with a standard laboratory instrument (such as a General Radio type 650-A impedance bridge or equivalent) shall be 85 ± 10 millihenries and the dc resistance shall be 30 ± 3 ohms. When the primary coil is excited at 23.5V, 400 Hz, the signal frequency of the secondary coils shall be 800 Hz and the impedance shall be between 400 and 700 ohms at an angle between 70° and 80°. The corresponding impedance of the secondary coils (leg-to-leg) shall be between 260 and 320 ohms at an angle between 72° and 88° at the 800-Hz signal frequency. The transmitter shall not become magnetized either by the presence of strong magnetic fields or a dc current passing through the secondary coils. The transmitter shall be designed to minimize crosstalk errors and to equalize the sensitivity coefficients for use with electrical compass swinging techniques.

3.5.1.1 Mechanism and Pre-Indexing - The transmitter mechanism sensor shall be mounted pendulously with respect to the case. The transmitter shall incorporate an interchangeable pre-indexing capability. The dimensions for mounting and pre-indexing the transmitter mechanisms with the pre-index mounting rings are given in Figure 2 and 2A. The penduluous element set forth in Paragraph 3.5.1.2 shall be adjustable in azimuth with respect to the mounting holes of Figure 2A and shall be adjusted so that the index error is less than 0 ± 0.1 degree when the transmitter mechanism is mounted in the fixture of Figure 4 prealigned to the local magnetic field. The pre-index mounting

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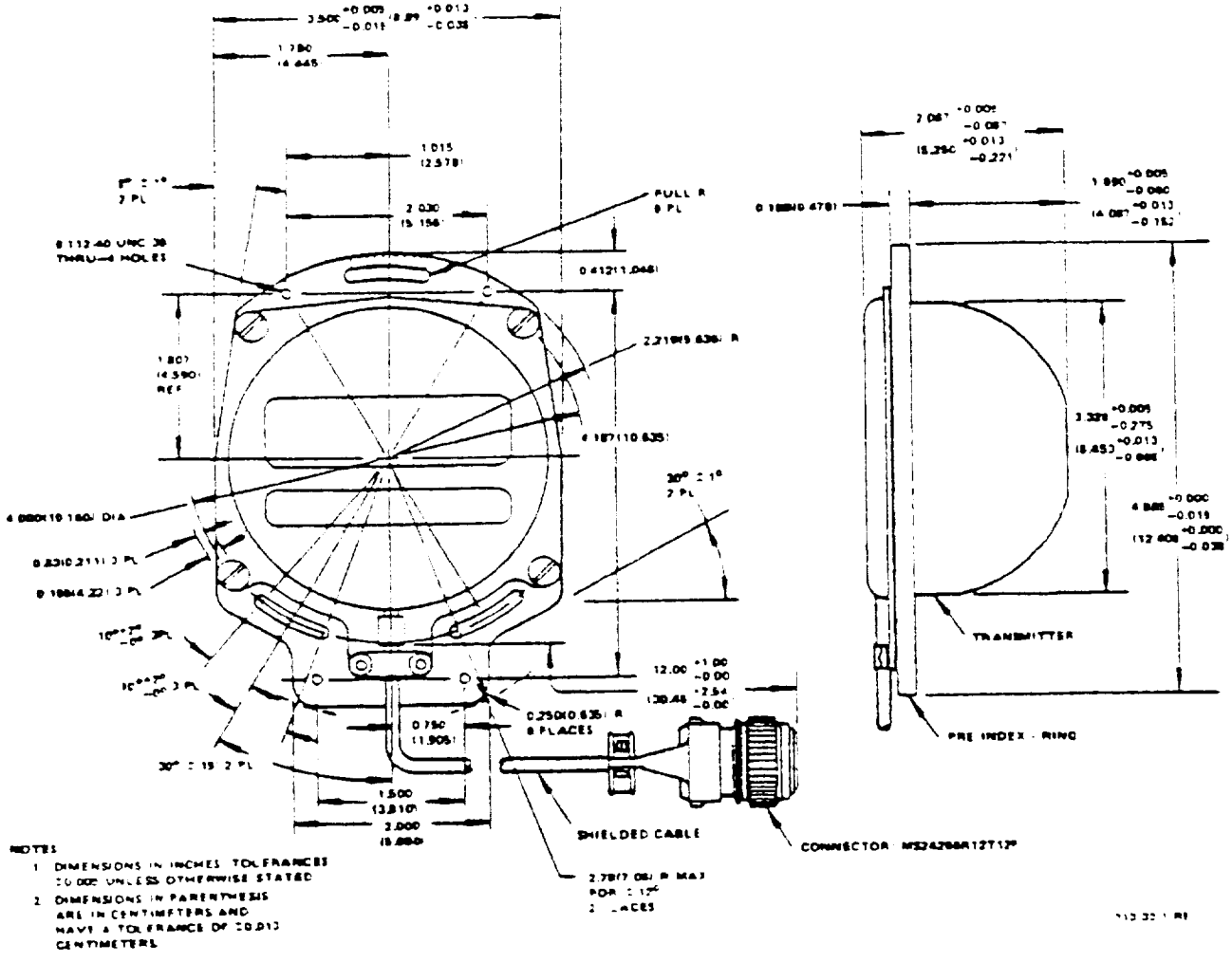


Figure 2
TRU-79A/A Transmitter

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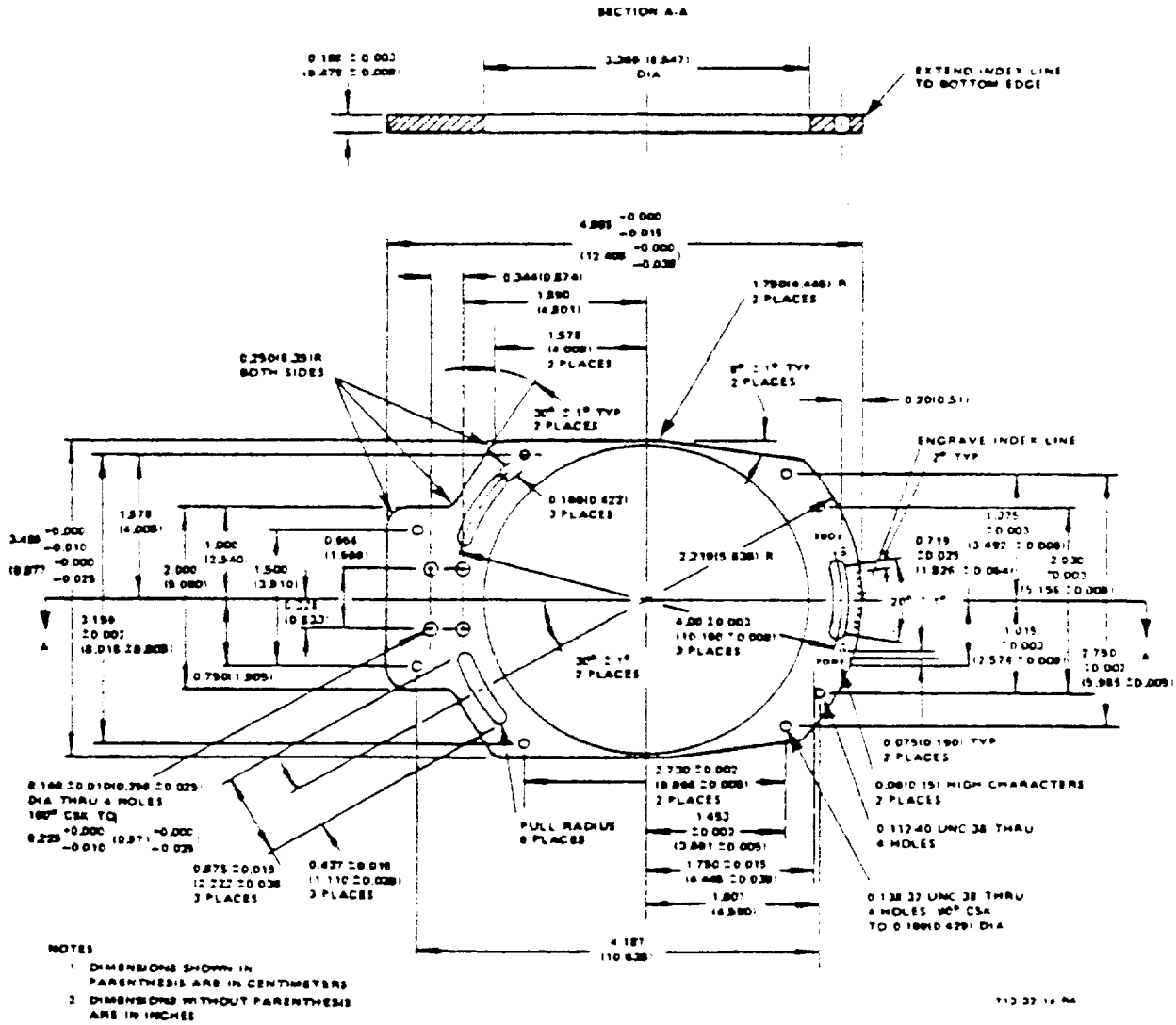


Figure 2A
Pre-Indexing Mounting Ring

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ring shall permit the interchange of any pendulous transmitter mechanism in a calibrated and compensated aircraft installation without creating an index error exceeding that specified in 4.6.25.

3.5.1.2 Pendulous Element - The earth's field inductor shall be mounted on a pendulous element so that it will tend to be horizontal when at rest. The pendulous element shall have at least 30° of freedom in pitch and at least 10° of freedom in roll with respect to the case. The pendulous element shall have a time period of not less than 0.2 second nor more than 0.75 second. The maximum permissible scale error shall be ±1.25° at a tilt angle of ±15° in pitch or ±10° in roll.

3.5.1.2.1 Damping - Fluid-type damping shall be used to stabilize the pendulous element. The damping system shall permit not more than 12 oscillations to occur when the pendulous element is made to swing from any position up to its maximum degree of freedom.

3.5.1.3 Sealing - The transmitter shall be hermetically sealed to prevent leakage of the damping fluid.

3.5.1.4 Core Material - The material for the magnetic core of the transmitter shall be such that it will withstand the maximum operating temperature specified herein plus the temperature rise due to heat dissipation from the electrical circuitry, without permanently altering the magnetic properties of the material.

3.5.1.5 Wire - Wire for the external electrical circuits of the transmitter shall be in accordance with MIL-W-22759 using silver-placed, Class I conductors except that insulation material shall be teflon.

3.5.1.6 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.5.1.7 Level Field - Variation of the vertical field from 0.54 to 0.02 to less than 0.02 oersted shall not change the scale error output of the transmitter by more than ±0.3°.

3.5.1.8 Signal Strength - When tested as specified in 4.6.10, the minimum 800-Hz signal shall not be less than 1000 mV.

3.5.1.9 Signal Harmonic Components - When tested as specified in 4.6.11, the maximum 400-Hz signal shall not exceed 25 mV and the maximum 1,200-Hz signal shall not exceed 40 mV.

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3.5.1.10 Accuracy - Accuracy of the transmitter under normal operating conditions and at temperatures from -30° to +150°C shall be within ±0.4° of true heading; between 150° and 200°C shall be within ±0.8° of true heading; and between -55° and -30° shall be within ±1.25° of true heading.

3.5.1.11 Sensitivity - With the transmitter set to the 0° or north heading and the signal transformer set to a heading of 90° ± 0.05° greater than the transmitter, a current application of 1.07 ± 0.05 mA from pins 1 to 6 shall result in a null indication on the signal transformer output.

3.5.1.12 Crosstalk Error (180°) - When the transmitter is set at the 0° or north heading, a current application of 2.14 ± 0.1 mA dc from pin 1 to 6 shall generate a 180° electrical heading. The error at this electrically simulated heading shall not differ from the corresponding manual scale error at 180° by more than ±0.8°.

3.5.1.13 Crosstalk Error (270°) - When the transmitter is set on the 90° heading, a current application 1.2 ± 0.1 mA dc from pin 2 to pin 3 shall generate a 270° electrical heading. The error at this electrically simulated heading shall not differ from the corresponding manual scale error at 270° by more than ±0.8°.

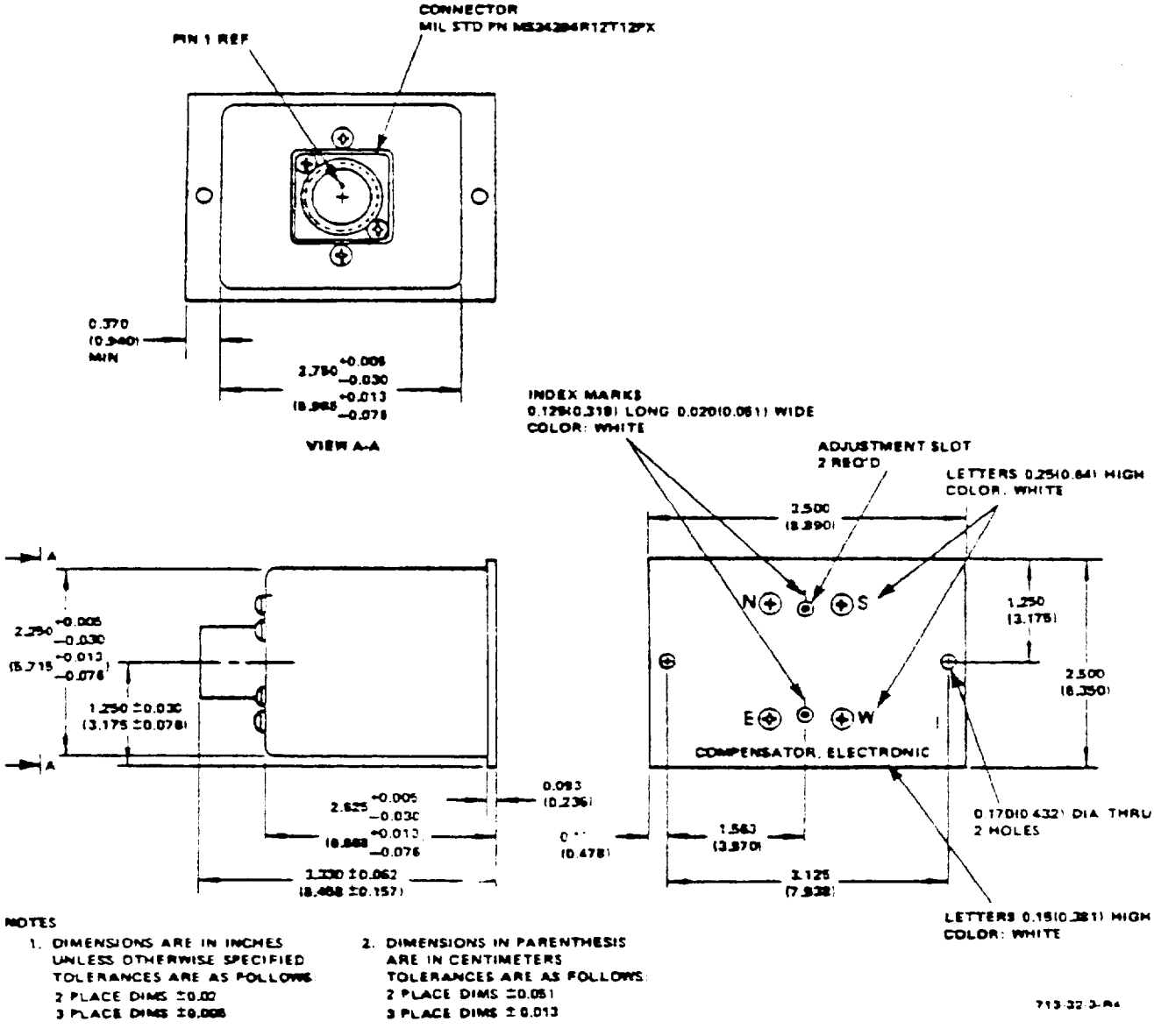
3.5.1.14 Electrical Connectors - The electrical connector provided on the transmitter shall be a MS24266R12T12P, located as shown in Figure 2 and in accordance with MIL-E-5400.

3.5.2 Compensator

3.5.2.1 General Design - The compensator shall be designed for mounting remotely from the transmitter. Interconnection between the compensator and transmitter shall be as shown on Figure 1. The degree of compensation specified herein shall apply for interconnecting lead lengths between the compensator and transmitter of up to 200 feet of No. 22 wire. The compensator shall incorporate two screwdriver slot-type adjustments by means of which it shall be possible to compensate for errors in the heading signal produced by the transmitter. One adjustment shall be capable of compensating for north-south transmitter errors and shall be labeled N-S and the other adjustment shall be capable of compensating for east-west transmitter errors and shall be labeled E-W. The compensator shall conform to figure 3.

3.5.2.2 Neutral Position Influence - With both the N-S and E-W adjustments set on their neutral positions, the compensator shall not change the heading signal produced by the transmitter by more than 1° upon removal or reinsertion of the compensator into the transmitter circuitry.

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NOTES

1. DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES ARE AS FOLLOWS 2 PLACE DIMS ±0.00 3 PLACE DIMS ±0.005
2. DIMENSIONS IN PARENTHESIS ARE IN CENTIMETERS TOLERANCES ARE AS FOLLOWS 2 PLACE DIMS ±0.051 3 PLACE DIMS ± 0.013

Figure 3
MXU-394/A Compensator

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3.5.2.3 Compensator Range - With the transmitter set on 0° heading, the N-S adjustment shall be capable of changing the output signal by ±4° to ±6° for the first ±360° rotation of the N-S adjustment from its neutral position. An additional 360° rotation of the N-S adjustment in the same direction shall deflect the output signal at least 9° from the neutral position. With the transmitter setting on 90°, the E-W heading adjustment shall be capable of changing the output signal by ±4° by ±6° for the first ±360° rotation of the E-W adjustment from the neutral position. An additional 360° rotation of the E-W adjustment in the same direction shall deflect the output signal to at least 9° from the neutral position. A clockwise rotation of each adjustment as specified above shall cause an increase in the resultant heading signal.

3.5.2.4 Compensator Adjustment Interference - A 360° rotation of the E-W adjustment screw shall not influence the transmitter output signal by more than ±0.5° when it is set at 0° heading. When set at 90° heading, a rotation of 360° on the N-S adjustment shall not influence the transmitter output signal by more than ±0.5°.

3.5.2.5 Electrical Connector - The electrical connector MS24264R12T12PNX for the compensator shall be located as shown in Figure 3.

3.6 Identification of Product - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.6.1 Part Numbering of Interchangeable Parts - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Workmanship - The detector, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.7.1 Dimensions and Tolerances - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the detector, they shall be held or limited accordingly.

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3.7.2 Screw Assemblies - Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.7.3 Cleaning - The detector shall be thoroughly cleaned of loose, spattered or excess solder, metal chips and other foreign material during and after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

3.7.4 Weight - The weight of the transmitter shall not exceed 28.8 ounces (818 gm). The weight of the compensator shall not exceed 7.5 ounces (213 gm).

3.7.5 Pivots, Bearings, and Gears - Pivots, bearings, and gears shall not bind in operation. These parts and parts necessary for electrical commutation (i.e., sliprings, brushes, and rotary contacts) shall be as nearly frictionless as possible except where friction is required.

3.8 Power

3.8.1 Primary Input Power Requirements - The detector shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704. Excitation for the primary circuit of the detector shall be 23.5 ± 2.0V, 400 ± 40-Hz single-phase ac power.

3.8.2 Output - The 800-Hz signal output from the transmitter secondary circuit shall be transmitted over a 3-wire line so that it may be connected to the stator of the signal transformer.

3.8.3 Maximum Output - The transmitter and signal transformer shall be placed on the 0° heading (null position) and the 800-Hz signal voltages across the 1-2 and 1-3 detector legs shall be recorded. The transmitter and signal transformer shall then be rotated to the 30° heading (null position). The voltage across the 1-2 leg shall increase and across the 1-3 leg shall decrease. With the transmitter still on the 30° heading, the signal transformr shall be rotated to the 120° heading. The line-to-line voltage across the 1-2 leg shall be 115 mV minimum.

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4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for Inspection and Test - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests - Items covered by the specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Qualification Tests

(2) Quality Conformance Tests

4.3 Qualification Tests - The qualification test shall consist of all the tests of 4.6 performed in the order listed and shall include the reliability test for the Reliability Qualification Test Phase.

4.3.1 Qualification Samples - Qualification inspection samples shall consist of six samples. The samples submitted shall have been previously subjected only to the individual tests. Three samples shall be forwarded at the contractor's expense to the laboratory designated in the letter of authorization. Three samples shall be retained by the contractor for the Reliability Qualification Phase, which will be run by the contractor at his plant.

4.3.1.1 Qualification Test Sample Identification - The qualification test samples shall be plainly identified by durable tags, securely attached, and marked with the following information.

Sample for Qualification Test
(Name of Equipment)
Submitted by (Manufacturer's name, date and part number)
for Qualification test in accordance with
Specification MIL-_____ under authorization
(reference letter authorizing tests.)

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4.4 Quality Conformance Test - The contractor shall furnish all samples and shall be responsible for accomplishing all tests except Sampling Plan B tests which will be conducted at a government laboratory designated by the procuring activity. Quality conformance test, except for Sampling Plan B, shall be under the supervision of the government quality control representative. The contractor shall furnish test reports showing detailed quantitative results for all tests required by this specification, signed by an authorized representative of the laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality Conformance tests shall consist of the following tests:

(a) Individual Tests

(b) Sampling Plan A Tests

(c) Sampling Plan B Tests

(d) Reliability Assurance Tests

(e) Longevity Tests

4.4.1 Individual Tests - Each instrument submitted for acceptance shall be subjected to the Individual Test. This test shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each instrument accepted shall have padded the following tests:

4.4.1.1 Transmitter -

Examination of Product
Scale Error
Sensitivity and Crosstalk Tests
Level Field
Tilt Error
Damping
Signal Strength
Signal Harmonic Component
Mangetization

4.4.1.2 Compensator -

Examination of Product
Neutral Position
Compensator Range
Interference

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4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B tests. The tests samples selected for sampling tests shall first have passed the Individual Test. The test samples which have been subjected to Sampling A test shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual Tests. Test samples which have been subjected to the Sampling Plan B test shall not be delivered on contract.

Quantity Offered For Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

When a defective instrument is detected, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the next fifteen and proceed as indicated. (One from next 50, etc)

4.4.2.1 Scope of Sampling Plan A Test - Each sample selected for Sampling Plan A Test shall be subjected to the following tests in the order listed:

4.4.2.1.1 Transmitter

Individual Tests
Voltage and Frequency Variation
Low Temperature
High Temperature
Temperature Altitude
Vibration Failure
Vibration Error

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4.4.2.1.2 Compensator

Individual Tests
Voltage and Frequency Variation
Low Temperature
High Temperature
Temperature Altitude
Vibration Failure
Vibration Error

4.4.2.2 Sampling Plan B Test - One instrument stall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded at the contractor's expense to a government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information.

(Name of Equipment)
Submitted by (Manufacturer's name, date)
for Production Acceptance Sampling Plan B
Test of MIL-_____ in accordance with
Contract/Order No. _____
Manufacturer's Part No. _____

4.4.2.2.1 Sampling Plan B Approval - Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the Contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Scope of Sampling Plan B Test - Each sample selected for Sampling Plan B Test shall be subjected to the following tests in the order listed:

Sampling Plan A Tests
Pre-Indexing Interchangeability
Dust
Acceleration
Explosive Atmosphere (compensator only)
Shock
Special Vibration (transmitter only)
Humidity
Fungus
Salt Fog

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4.4.3 Reliability Assurance Test - Reliability assurance tests shall be conducted using MIL-STD-781. Test as required by both the qualification phase and Production Acceptance (Sampling) Phase shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under this contract, a minimum of 3 detectors specified in 1.2 shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase for Production Reliability Tests". For the Qualification Phase, Test Level J, Test Plan III shall be used except as follows:

(a) Temperatures: Temperatures for the compensator shall range from -54° to +71°C.

(b) In addition to the vibration specified, the transmitter shall be subjected to a 1.0-inch double-amplitude vibration at 10 Hz ± 2 Hz for 50 minutes after every 100 hours of operation.

The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.1.1 Performance Tests - At the completion of every 250 ± 10 hours of operation, the detector shall be subjected to the tests specified in Paragraph 4.6.2. At the completion of 500 ± 20 hours of operation and at the conclusion of this test, the detector shall be subjected to all the Individual Tests of Paragraph 4.6.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781B (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". The test level shall be in accordance with Paragraph 4.4.3.1.

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot. The Accept-Reject criteria for Test Plan V shall be used to determine the length of the test in each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

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4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc, shall be a part of the test procedures to be submitted by the contractor and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design detect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the Individual Test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4.6 Longevity Tests - Longevity Tests shall be performed on at least two detectors in accordance with Test Plan XXVIII of MIL-STD-781. The contractor shall furnish all samples and shall be responsible for accomplishing the longevity test. Two of the detectors that have completed the reliability tests of Paragraph 4.4.3 shall continue to be tested for a total of 10,000 hours (5,000 hours each). Test time accumulated on the detector during the reliability test may be applied toward the longevity test.

4.4.6.1 Test conditions - Test conditions shall be in accordance with Test Level J of MIL-STD-781 except as follows:

(a) Temperatures for the compensator shall range from -54° to +71°C.

(b) No accept/reject criteria for this test are specified.

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4.4.6.2 Performance Check - At the completion of 500 ± 20 hours of operation and at the conclusion of this test, the detector shall be subjected to all the Individual Tests of Paragraph 4.6. A visual check shall be made daily to ensure the detector is functional.

4.4.6.3 Test Data - The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(a) Part type number.

(b) The circuit reference symbol number.

(c) The part function.

(d) Name of the manufacturer.

(e) Nature of the failure.

(f) The number of hours which the part operated prior to failure.

4.4.6.4 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether that are acceptable.

4.4.6.5 Reconditioning of Longevity Test Samples - An equipment which has been subjected to the longevity test shall be reconditioned as follows:

(a) On completion of the longevity test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

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4.5 Standard Conditions for Test

4.5.1 Standard Atmospheric Conditions - Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C).

4.5.2 Attitude - Unless otherwise specified, the detector shall be tested in its normal operating position.

4.5.3 Vibration - Unless otherwise specified, the transmitter shall be subjected to vibration at a frequency from 24 to 35 Hz so that a point on the transmitter will describe, in a vertical plane, a circle with a diameter of not less than 0.002 (0.05 mm) nor more than 0.005 inch (0.13 mm).

4.5.4 Supply Voltage - Unless otherwise specified, all tests shall be conducted with 23.5 ± 0.5V, 400 ± 5-Hz single-phase ac power applied to the detector.

4.5.5 Field Strength - Unless otherwise specified, all tests shall be conducted with a horizontal field strength of 0.18 ± 0.02 oersted and a vertical field strength of 0.54 ± 0.02 oersted.

4.5.5.1 When a standard level field is required, the horizontal field strength shall be 0.18 ± 0.02 oersted and the vertical field strength shall be less than 0.02 oersted.

4.5.5.2 When an ambient level field is required, the horizontal field strength shall be the surrounding magnetic field at the test location and the vertical field strength shall be less than 0.02 oersted.

4.5.6 Standard Signal Transformer - All tests shall be conducted with a standard signal transformer having the following characteristics:

4.5.6.1 The signal transformer shall consist of a three-phase, three-wire stator surrounding a single-phase two-wire rotor. The maximum input line-to-line impedance of the stator with the rotor circuit open shall be 150 ± 15 ohms at a phase angle of 82° ± 3° when measured at 1,000 Hz with a standard laboratory instrument, such as a General Radio type 650-A impedance bridge, or equivalent.

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4.5.6.2 The rotor impedance with the stator open shall be 10,000 ± 1,000 ohms at a phase angle of 85° ± 2°. When the rotor is tuned for maximum 800 Hz signal and 90 ± 2 mV, 800 ± 5 Hz is applied between one stator lead and a bus connecting the other two stator leads. The maximum 800-Hz signal from the rotor shall be 900 ± 90 mV. The rotor shall be rotated until maximum voltage is obtained.

4.5.6.3 The signal transformer errors shall be minimum and known so that the error can be removed from the heading error during testing of the detector.

4.5.6.4 The signal transformer shall be suitably mounted so that angular displacement from 0° to 360° may be determined within ±0.1°.

4.5.7 Turntable and Test Fixture - All tests, except where specified, shall be conducted, with the pre-indexing ring removed, on a turntable equipped with a fixture conforming to Figure 4. The turntable shall provide accurate 15-degree increments from 0 degree to 360 degrees. Before a test is conducted a master compass or master transmitter shall be placed on the test fixture with the turntable aligned to 0 degree. The fixture shall then be adjusted with respect to the turntable until the fixture is aligned with the local magnetic field.

4.5.8 Standard Null Meter - The standard null meter shall have a minimum input impedance of 1 megohm and shall be capable of detecting the null position of the standard signal transformer to within ± 0.5° when the detector is in a horizontal field strength of 0.08 oersted and zero vertical field strength.

4.5.9 Remote Compensator - Unless otherwise specified, the remove compensator shall be connected into the transmitter circuitry during all testing. This is for convenience of testing and does not compromise the requirement for interchangeability of components.

4.5.10 Magnetic Heading Simulator (MHS) - A MHS may be used in lieu of the transmitter in the tests specified in 4.6.4 through 4.6.6. The MHS shall consist of a transmitter sensing element mounted in a double magnetic shield enclosure and circuitry capable of controlling a magnetic field within the enclosure. The MHS shall produce accurate and stable simulated magnetic heading in 15° increments from 0° to 360°.

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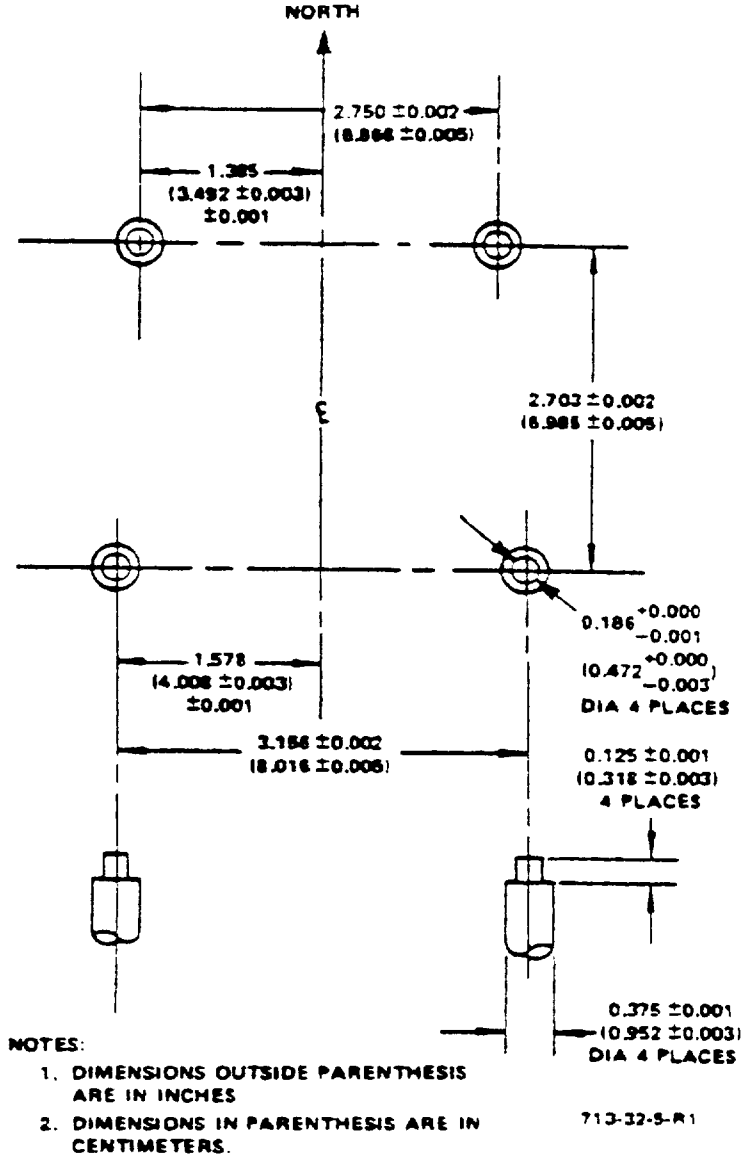


Figure 4
Test Fixture Details

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4.5.11 Wave Analyzer - The wave analyzer used in testing the transmitter shall be a Hewlett Packard Model 302A or equivalent.

4.6 Test Methods

4.6.1 Examination of Produce - The detector shall be examined to determine compliance with the requirements specified herein with respect to materials, design and construction, mechanical measurements, workmanship, and marking.

4.6.2 Scale and Index Error - The transmitter mechanism with the pre-indexing mounting ring removed shall be placed on the aligned test fixture described in Paragraph 4.5.7. The field strength shall conform to Paragraph 4.5.5. and the excitation shall be 23.5 volts, 400 Hz. The transmitter secondary circuit shall be connected to the stator of a standard signal transformer whose rotor is connected to a standard null meter. The transmitter and signal transformer shall be set at the 0 degree or North heading (rotor at electric null position). The transmitter shall then be turned in any heading increment from 0 degree through 360 degrees. Each selected increment shall be less than 30 degrees. At each selected heading, the pointer of the signal transformer shall be turned correspondingly to its null position and the reading taken of the pointer on the dial. The algebraic average of the recorded errors between 0 degree and 360 degrees is the index error and it shall not exceed 0 ± 0.2 degree. No single reading shall exceed 0 ± 0.4 degree after the index error has been subtracted.

4.6.3 Sensitivity and Crosstalk Tests - The following tests shall be performed with the transmitter connected to the signal transformer as specified in 4.6.2.

4.6.3.1 Sensitivity - The transmitter shall be set to the 0 or North heading and the signal transformer set to a heading of 90° ± 0.05° greater than the transmitter. A direct current shall be applied from pins 1 to 6 and varied until a null is indicated on the standrad null meter. The direct current snall be 1.07 ± 0.05 mA. This test shall be performed in the standard level field.

4.6.3.2 Crosstalk (180°) - The transmitter shall be set at the 0° heading and a direct current of 2.14 ± 0.1 mA shall be applied from pins 1 to 6. This shall produce a 180° heading in the transmitter. The error at this heading shall not differ from the error obtained in a scale error test at the 180° heading by more than ±0.8° when corrected for misalignment of the transmitter and readout. This test shall be performed in the ambient level field.

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4.6.3.3 Crosstalk (270°) - The transmitter shall be set at a 90° heading and a direct current of 1.2 ± 0.1 mA shall be applied from pin 3 to pin 2. This shall produce a 270° heading in the transmitter. The error at this heading shall not differ from the error obtained in the scale error test at the 270° heading by more than ±0.8° when corrected for misalignment of the transmitter and readout. This test shall be performed in the ambient level field.

4.6.4 Neutral Position - The transmitter shall be set at the 0° heading as specified in 4.6.2. The compensator, with both north-south and east-west screws set at neutral, shall then be connected to the transmitter circuitry. The resulting shift of the null position of the signal transformer rotor shall not exceed ±1°. This test shall be repeated with the transmitter set on the 90° heading.

4.6.5 Compensator Range - The transmitter shall be set at the 0° heading as specified in 4.6.2. The compensator, with both north-south and east-west screws set at neutral, shall be connected to the transmitter circuitry. The north-south screw shall then be turned 360°. The shift in null position of the rotor shall be between 4° and 6°, plus or minum, depending on the direction of rotation of the adjustment screw. The same screw shall then be turned an additional 360° in the same direction. The resulting shift in null position of the rotor shall be at least 9° from the neutral position. This test shall be repeated with the transmitter set on the 90° heading using the east-west screw and leaving the north-south screw at neutral. The corresponding shift in null position shall be within the limits specified herein for the north-south adjustment range.

4.6.6 Interference - The transmitter shall be set at the 0° heading as specified in 4.6.2. The compensator shall then be connected to the transmitter circuitry with the north-south screw set at neutral and the east-west screw set at 360° from neutral. The resulting shift in null position of the rotor shall not exceed ±0.5°. This test shall be repeated with the transmitter set on the 90° heading, the east-west screw set at neutral, and with the north-south screw set 360° from the neutral position. The resulting shift in null position of the rotor shall not exceed ±0.5°.

4.6.7 Level Field - The transmitter shall be connected to the signal transformer as specified in 4.6.2 on the 0° heading. The horizontal field strength shall be 0.18 oersted and the vertical field shall be less than 0.02 oersted. Heading errors shall be recorded in increments of 60° through 360°. The difference between these errors, after index error has been subtracted, and the corresponding headings in 4.6.2 shall not exceed ±0.3°.

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4.6.8 Tilt Error - The transmitter shall be connected to the signal transformer as specified in 4.6.2 on the 0° heading. The transmitter shall be tilted up 15° in pitch and a scale error reading taken. This test shall be repeated with the transmitter tilted down 15° in pitch, and then at a tilt of ±10° in roll. The error between the signal transformer rotor and the transmitter shall not exceed 1.25° at the above test points after the index error has been subtracted.

4.6.9 Damping - The transmitter shall be tilted rapidly from the level position to 10° in roll and returned to the level position. The null position of the standard signal transmitter shall not oscillate more than 0.1° after 12 oscillations. This test shall be repeated except the transmitter shall be tilted 15° in pitch.

4.6.10 Signal Strength - The detector shall be connected to the signal transformer as specified in the scale error test on the 0° heading. The signal transformer then shall be rotated to its 90° heading and the 800-Hz voltage across the rotor leads recorded. The voltage shall be not less than 1,000 mV.

4.6.11 Signal Harmonic Components - The detector shall be connected to the signal transformer as specified in the scale error test. The signal transformer shall be rotated for a maximum 400-Hz signal which shall not exceed 25 mV. It shall then be rotated for a maximum 1,200-Hz signal which shall not exceed 40 mV.

4.6.12 Magnetization - The transmitter shall be set at the 0° heading with field strengths as specified in the level field test. A dc current of 100 ± 10 mA shall be passed through the transmitter secondary coils in turn, 1 to 6, 2 to 6, and 3 to 6. Heading errors shall be recorded in increments of 60°. The difference between these errors, when corrected for index error, and the corresponding errors obtained in the level field test shall not exceed 0.2°. This test shall be repeated with the dc current passed through the secondary coils in the 6 to 1, 6 to 2, and 6 to 3 direction.

4.6.13 Voltage and Frequency Variation - The detector shall be set up as specified in the scale error test with the signal transformer on the 0° heading. The supply voltage shall then be reduced to 22V and the frequency to 360 Hz. The shift in null position of the signal transformer rotor shall not exceed ±0.5°. This test shall be repeated on two other headings not less than 100° apart. The supply voltage shall then be increased to 25V and the frequency to 440 Hz. The same test limits shall apply.

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4.6.14 Low Temperature - While inoperative, the detector shall be subjected to a temperature of -55 °C for a period of 4 hours. At the end of this time and while at the specified temperature, it shall be energized and subjected to the scale error test, except that the tolerance shall be ±1.25°. The transmitter shall show no signs of leakage and the compensator shall operate without excessive binding or excessive freedom. The detector shall then be returned to room temperature. The detector shall meet the requirements of the Individual Tests.

4.6.15 High Temperature - The transmitter and compensator shall be excited for a period of 4 hours at ambient temperatures of 200° and 71° C, respectively. At the end of the 4-hour period and while at the specified temperatures, they shall be subjected to the scale error test except that the tolerance shall be ±0.8°. The transmitter shall show no evidence of leakage and the compensator shall operate without excessive binding or freedom. The transmitter and compensator shall then be returned to room temperature and subjected to the requirements of the Individual Tests.

4.6.16 Temperature-Altitude - The temperature-altitude test shall be conducted as follows:

(a) Step 1: While operating, the detector shall be subjected to a pressure equivalent to 80,000 ± 1,000 feet at a temperature of -55° ± 2 °C for a continuous period of 3 hours. At the end of the 3-hour period and still at this temperature and pressure the detector shall be subjected to the scale error test. The permissible scale error tolerance shall not exceed ±1.25° of true heading. This test shall be conducted on the 0° heading and two other headings not less than 100° apart. At the end of the 3-hour period, the detector shall show no sign of leakage nor shall there be any damage that would affect subsequent operation.

(b) Step 2: The compensator only shall be removed from the chamber. A suitable connection between the transmitter and compensator shall be made through the chamber wall. The transmitter shall then be stabilized at a temperature of 160° ± 2 °C, the altitude adjusted to 60,000 ± 1,000 feet and this condition maintained for a continuous period of 3 hours. At the end of the 3-hour period and still at this temperature and pressure, the scale error test shall be performed as in step 1 except the tolerance shall be ±0.8° of true heading. At the end of the 3-hour period, there shall be not evidence of leakage or damage that would affect subsequent operation. The detector shall be returned to room temperature and shall meet the requirements of the Individual Tests.

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4.6.17 Vibration

4.6.17.1 Vibration Failure (Compensator) - The compensator shall be subjected to a vibration test in accordance with MIL-T-5422, Procedure 1, Figure 6, Curve 111a. At the completion of this test, the compensator settings shall not be changed nor shall there be any looseness or damage. The compensator shall then be subjected to the Individual Tests.

4.6.17.2 Vibration Failure (Transmitter) - While operating, the transmitter shall be subjected to a vibration test in accordance with MIL-T-5422, Procedure II, Part II, except the vibration curve shall be in accordance with Figure 5. Performance test while the detector is mounted on the vibrator shall not be required. At the completion of this test, the transmitter shall be subjected to and shall meet the requirements of the Individual Tests. The transmitter shall show no evidence of leakage.

4.6.17.2.1 Special Vibration Test - The transmitter shall be vibrated at a frequency of 900 cpm with a total excursion of 0.125 inch (3.18 mm) in a horizontal plane for a period of 100 hours. The transmitter shall be mounted on the vibration machine in such a manner that the zero index line on the transmitter flange will be 45° from the direction of the vibration machine motion. At the completion of 100 hours, the transmitter shall meet the requirements of the Individual Tests.

4.6.17.3 Vibration Error - While operating, the detector shall be subjected to a vibration error test in accordance with MIL-T-5422, Figure 5, Curve III, except the following test periods shall apply:

(a) Resonance Dwell - none

(b) Sinusoidal Cycling - 1 hour

(c) Sweep Time - 20 minutes

During this test, the transmitter and compensator shall be interconnected. During vibration, the scale error shall be measured on the 0° heading and on two other headings not less than 100° apart. Scale error shall not exceed 1° of true heading.

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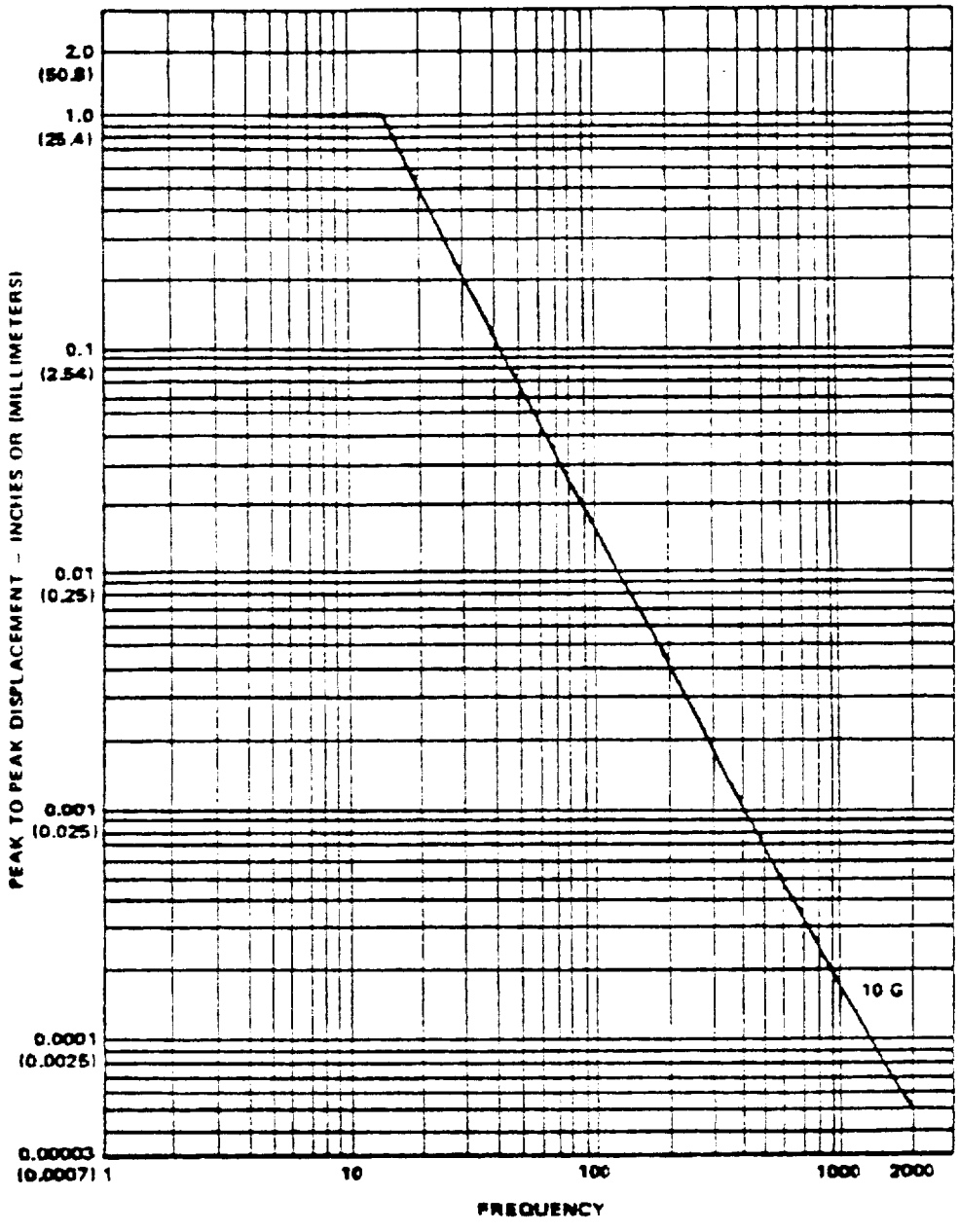


FIGURE 4 VIBRATION 713-32-4-R2

Figure 5
Vibration Failure (Transmitter)

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4.6.18 Dust - The detector shall be subjected to a dust test in accordance with MIL-T-5422. At the end of this test, the detector shall be removed from the test chamber and allowed to cool at room temperature. The detector shall not be damaged. The compensator adjusting screws shall operate without excessive looseness. The detector shall then be subjected to and shall meet the requirements of the Individual Tests.

4.6.19 Acceleration - The detector with power applied shall be subjected to an acceleration test at the levels set forth in Paragraph 3.3.8.6 for a period of 1 minute in each direction. The detector shall then be subjected to and shall meet the requirements of the Individual Tests.

4.6.20 Explosive Atmosphere - The compensator shall be subjected to an explosion-proof test in accordance with MIL-T-5422, procedure 1.

4.6.21 Shock - The detector shall be subjected to a shock test in accordance with MIL-T-5422. Performance test while the detector is subjected to shock testing shall not be required. At the completion of this test the detector shall show no signs of leakage or damage that would affect subsequent operation. The detector shall then be subjected to and shall meet the requirements of the Individual Tests.

4.6.22 Humidity - The detector shall be subjected to a humidity test in accordance with MIL-T-5422. The detector shall then be subjected to and shall meet the requirements of the Individual Tests. There shall be no evidence of corrosion or rust of internal or external parts.

4.6.23 Fungus - The detector shall be subjected to a fungus test in accordance with MIL-T-5422. At the conclusion of this test, the detector shall be subjected to and shall meet the requirements of the Individual Tests. There shall be no evidence of deterioration that might adversely affect subsequent operation nor shall fungus growth be supported.

4.6.24 Salt Fog - The detector shall be subjected to a salt fog test in accordance with MIL-T-5422. At the conclusion of this test, the detector shall be subjected to and shall meet the requirements of the Individual Tests. The compensator adjusting screws shall operate without excessive sticking, binding, or looseness. There shall be no evidence of corrosion of internal or external parts.

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4.6.25 Pre-Indexing Interchangeability - Place any transmitter's pre-indexing mounting ring on the turntable of 4.5.7 and align to 0 degree using a master transmitter. Place any transmitter in the pre-indexing mounting ring and perform a scale error test in accordance with 4.6.2. The resulting index error shall not change by more than 0 ± 0.1 degree from the initial index error test of 4.6.2 performed on that transmitter.

4.7 Inspection of Preparation for Delivery - Preservation, packaging, and packing shall be inspected to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A (reference definition in MIL-STD-794) - Unless otherwise specified in the contract or order, individual components of the detector shall be packaged one each in accordance with method IIC of MIL-P-116. Overbox complete pack in PPP-B-636 container utilizing cushioning of a type, density, configuration and thickness to ensure shock experienced by item does not exceed peak values established for the item, specified in g's when completed packs are submitted to the rough handling tests of MIL-P-116.

5.1.2 Level C - Unless otherwise specified by the contract or order, individual components of the detector shall be packaged identically to level A requirements stipulated in 5.1.1 except method III of MIL-P-116 shall apply.

5.1.3 Magnetic Shielding - Shielding as required shall be applied in accordance with MIL-S-4473.

5.2 Packing - Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A - Detectors preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds (90.91 kg). Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

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5.2.2 Level B - Detectors preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.3 Level C - Packing shall be applied which affords adequate protection during direct domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

6. NOTES

6.1 Intented Use - The DSU-4A/A detector covered by this specification is intended for use in airborne heading systems to detect the direction of the earth's magnetic field with respect to the direction of the aircraft.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title number and date of this specification

(b) When sampling Plan B test will not be conducted

(c) Levels of packaging and packing required

(d) Compass calibrator compatibility requirement

(e) Maintainability demonstration requirement

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

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(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Hermetic Seal - Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes brazing or soldering, or fusion by welding. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings

6.5 Associated Equipment - The detector shall operate with the following associated equipment:

(a) Attitude and Heading Reference System, type A/A24G-27 and -39

(b) Attitude and Heading Reference System, type AN/ASN-50

(c) Attitude and Heading Reference System, type AN/ASN-75

(d) Attitude and Heading Reference System, type AN/ASN-110

(e) Attitude and Heading Reference System, type AN/ASN-111

(f) Attitude and Heading Reference System, type AN/ASN-116

(g) Heading Reference System, type MA-1

6.6 Revision - Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.7 Intended Calibration Requirements - The TRU-79A/A Induction Transmitter is intended for future use with the MC-100 Magnetic Compass Calibrator Set, Sperry Part No. 4021608-901. K values shall be determined and permanently marked on the nameplate to relate the transmitter parameters to those of the MC-100.

6.8 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. Suppliers are directed to provisions governing qualification (SD-6) and are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification, in order to be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360 and information pertaining to qualification of products may be obtained from the activity.

6.9 Jurisdiction - This specification is under the cognizance of AIR-53354B.

Custodian:
Navy-AS

Preparing Activity
(Navy-AS)

Project No. 6605-N286

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-D-0038134B(AS) Dectector, Magnetic Azimuth DSU-4A/A, High Temperature

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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